This assignment will not be graded and is only for practice.

Find the dimension of the vector space

$$V = \{(x, y, z) \mid x, y, z \in \mathbb{R}, x + y + z = 0, z = 0\}$$
$$\textit{Addition: } (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2);$$
$$(x_1, y_1, z_1), (x_2, y_2, z_2) \in V$$
$$\textit{Scalar multiplication: } c(x, y, z) = (cx, cy, cz);\ (x, y, z) \in V,\ c \in \mathbb{R}$$

Solution:

Before we proceed to the solution you may want to recall the following weekly contents:

- Watch : Week 2 Lecture 6, Week 3 Lecture 3 and 4, Week 4 lecture 1 and 2, Tutorial 3
- Solve : Activity 4.1 Question 9 and Activity 4.2 Question 6, Activity 4.3 Question 7
- Solve : Practice assignment: Question 2.

Observe that, in question, you are asked to find the dimension of the vector space . So first you have to find a basis of the vector space.

Note: Consider the system of linear equations

$$x + y + z = 0$$
$$z = 0$$

Find the solution of the system of linear equations.

Step 1:

1) The system of linear equations has **1 point**

- unique solution.
- no solution.
- infinitely many solution.

No, the answer is incorrect.
Score: 0

Feedback:

Use Gauss elimination method, and observe that $z = 0$ and $x = -y$.

Accepted Answers:

infinitely many solution.

2) Is (1,-1,0) is a solution of the system of linear equations? **1 point**

- Yes
- No

No, the answer is incorrect.
Score: 0

Feedback:

Substitute $x = 1, y = -1$ and $z = 0$ in both the equations of system of linear equations and check.

Accepted Answers:

Yes

3) Is $(1, -1, 0) \in V$? **1 point**

- ◯ Yes
- ◯ No

No, the answer is incorrect.
Score: 0

Accepted Answers:

Yes

Claim: $\{(1, -1, 0)\}$ spans V.
Step 2:

4) Let $(a, b, c) \in V$. Which of the following options is/are true? **1 point**

- ☐ $a = -b$
- ☐ $a + b = 1$
- ☐ $c = 0$
- ☐ $c = 1$

No, the answer is incorrect.
Score: 0

Feedback:

As vector (a, b, c) is a vector of the vector space then it will satisfy the system of linear equations $a + b + c = 0, c = 0$

Accepted Answers:

$a = -b$
$c = 0$

Hence, any vector $(a, b, c) \in V$ can be written as $(a, -a, 0)$.

5) Is $(a, -a, 0)$ a scalar multiple of $(1, -1, 0)$? ***1 point***

- Yes
- No

No, the answer is incorrect.
Score: 0

Accepted Answers:

Yes

6) If $(a, -a, 0) = \alpha(1, -1, 0)$, then what is the possible value for α? ***1 point***

- a
- $-a$
- 0

No, the answer is incorrect.
Score: 0

Accepted Answers:

a

Hence, the set $\{(1, -1, 0)\}$ spans the vector space V.

Step 3: In this step we have to check whether the set $\{(1, -1, 0)\}$ is linearly independent or not.

7) Let S be a set which contains only one non-zero vector. Then which of the following options is true? ***1 point***

- S is linearly independent set.
- S is linearly dependent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

S is linearly independent set.

Recall: A basis of a vector space is a set of linearly independent vectors which spans the vector space.

From here you can conclude that $\{(1, -1, 0)\}$ is the basis of the vector space.

Step 4: In this step we have to find the dimension the vector space V.

Recall: The cardinality of the basis is the dimension of V.

8) Find the dimension of the vector space V.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

Your score is: 0/8